

P001- Business Requirements Document

Summary	
Business Owner (SME):	Project Research Office
Project Manager (DT):	Aizhan Ismailova
Data Refresh Frequency	Daily
Requested on	18-Nov-2025
Estimated Effort :	6 weeks
Azure DevOps Ticket:	n/a
Status	In Progress

Project Summary

Section	Description
Executive summary	This dashboard provides the Project Office with a centralized and interactive view of research activities, including projects, patents, and publications. By integrating data from multiple domains into a unified analytical model, the dashboard enables management to assess project performance, funding utilization, researcher involvement, and overall research output. The solution supports data-informed decision-making and streamlines reporting processes that are currently manual and time-consuming.

Business objectives	• Provide visibility into ongoing and completed research projects across colleges and partner types. • Monitor funding allocation vs. actual utilization for each project. • Track researcher participation and identify workload distribution. • Consolidate key research outputs (publications, patents) to support institutional performance reporting. • Enable leadership to evaluate efficiency, progress, and contribution of research activities. • Reduce manual reporting effort and ensure consistent KPI measurement.
Background	The Project Office currently manages multiple independent datasets related to research activities, including project contracts, staff information, patents, and publications. These datasets exist in separate Excel files and manual trackers, creating inconsistencies and delays in reporting. A unified BI solution is required to standardize data definitions, automate KPI calculations, and deliver consolidated insights across all research domains. The implemented model uses conformed dimensions—particularly Staff—to integrate data across projects, patents, and publications.

Scope	In Scope: • Visualization of all research projects with details such as partner, start/end dates, domain, status, and funding. • Funding analysis, including contract amount and funding used. • Staff-level analytics: researcher demographics, college affiliation, employment type, and involvement in projects. • High-level KPIs for publications and patents linked to the same staff. • Automated data transformations (e.g., split Inventors, distribute revenue). • Interactive Tableau dashboard with filtering by college, status, partner type, and date ranges. Out of Scope: • Forecasting or predictive analytics. • Financial auditing or contract validation. • Data entry, operational workflows, or project management tools. • External system integration beyond provided datasets
-------	---

Features	● Funding Utilization Module: Contract amount vs. actual funding used, with variance indicators. ● KPI-s of totals of each Activity ● Trend Line of revenue ● Research Output Quality -
Limitations & Risks	● Data Quality Issues: Missing revenue, incomplete publication details, and inconsistent naming conventions may impact accuracy. ● Manual Source Files: Since source data is manually maintained, updates may not be timely or standardized. ● Revenue Distribution Logic: Dividing patent revenue among inventors may not reflect actual attribution unless validated by finance. ● Historical Gaps: Older projects or publications may not be fully captured due to incomplete archival data. ● Staff ID Dependency: Incorrect or missing Staff_ID values will break relationships between fact tables. ● No Real-Time Updates: Data is refreshed periodically; not suitable for operational, real-time decision-making. ● .

Data Source

university_test_dataset_large_messy.xlsx

Functional Requirements:

Dashboard Filters

FR-1 Filters:

Below filters are available for the end users to drill down/filter the dashboard.

1. Date filter (from/to) : Contains 2011-2025 to filter.
2. Domain. Contains list of industries where Projects were implemented
3. Status: project status

FR-2: BANS and KPIS:

The Tableau dashboard provides a unified view of:

- **Research Portfolio Overview:**

A high-level snapshot of total projects, publications, patents, commercialization rate, and total revenue generated, giving leadership instant visibility into the scale and value of ongoing research activities.

- **Funding Efficiency Analysis:**

A variance-based bar chart comparing actual funding used against contract amounts to highlight over-budget areas and identify domains that require financial attention.

- **Research Output Quality:**

A bubble chart plotting Impact Factor against Citations by college, shifting the focus from research quantity to research quality and helping assess which academic units contribute strongest to institutional ranking.

- **Cross-Domain Research Performance:**

Integrated visibility into projects, publications, patents, and staff involvement to provide a comprehensive, data-driven understanding of research productivity and impact across the university.

Non-Functional Requirements:

NFR-1: Performance

- Dashboard response time shall be under 5 seconds for standard filters.
- Queries generated from Tableau shall retrieve only necessary fields to minimize scan volume.
- Fact tables shall support efficient joins through proper indexing and clustering.

NFR-2: Scalability

- The system shall support expansion to additional research-related processes (e.g., grants, collaborations).

- Warehouse and data model architecture shall scale to accommodate increased volumes across academic years.

NFR-3: Data Quality

- Data validation rules shall enforce correct data types and remove inconsistencies during ingestion.
- All fact tables shall maintain referential integrity with their associated dimensions.
- Duplicate entries and malformed fields shall be automatically flagged.

NFR-4: Reliability & Consistency

- Foreign key constraints shall ensure valid Staff_ID, ProjectID, PatentID, and Publication_ID references.
- The model must support consistent KPI computation across all reporting periods.

NFR-5: Maintainability

- SQL scripts, transformation logic, and schema definitions shall be modular and documented.
- The dimensional model shall avoid redundancy to simplify long-term updates.

NFR-6: Security & Access Control

- Access to sensitive fields such as revenue (RevenueCorrect) shall be restricted to authorized users.
- Schema objects and data access privileges shall conform to organizational data governance standards.

NFR-7: Usability

- Dashboard layout shall be intuitive, with clear labeling, contextual KPIs, and consistent formatting.
- Filters shall be easy to use and support guided analysis for non-technical stakeholders.

NFR-8: Accuracy

- Calculations for KPIs (funding %, commercialization %, publication impact, etc.) shall reflect the approved business logic with no manual adjustments.

Sing off though Docusign